

MRM Framework: Multi-Source Statistical Foundation for Canadian Carceral, Police, and Oversight Data

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Abstract

The MRM (McNamara–Ruhela–Medina) framework provides a unified statistical foundation for analysing Canadian carceral, police, and oversight data by integrating five distinct data sources under a single set of estimators and identification primitives. The five sources span (i) the Ontario Tracking Information System (OTIS, A01-RCDD) provincial restrictive-confinement microdata, (ii) the federal Structured Intervention Unit Implementation Advisory Panel (IAP) reports together with independent academic replications, (iii) the Ontario Special Investigations Unit (SIU) police-oversight reports, (iv) the Toronto Police Service (TPS) open-data crime corpus and the Statistics Canada Crime Severity Index, and (v) the federal Corrections and Conditional Release Statistical Overview (CCRSO). The framework specifies a ten-estimator per-individual ensemble (inverse-probability weighting, augmented inverse-probability weighting, double-machine-learning interactive regression, propensity-score matching, partial-linear, and partial-linear super-learner variants), a generalised-estimating-equation clustering grid for high-cardinality groupings, a χ^2 family for aggregate contingency tables, a Mandela Rules classifier that operates at both the federal and provincial level, and a self-exciting Hawkes process for spatio-temporal modelling of TPS open-data events. The framework is implemented in the open-source MORIE toolkit as a curated set of MRM modules (Ruhela 2026c,b). Empirical replication of all published χ^2 statistics from the Sprott–Doob–Iftene academic record is demonstrated to within 0.01, and the provincial Ontario Mandela “torture”-classified proportion is reported to rise from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU rate of 9.9% documented in the published record by 10.7 percentage points at peak.

Keywords: causal inference, observational inference, Canadian carceral data, OTIS, Special Investigations Unit, Hawkes process, Mandela Rules, MRM framework, sociolegal statistics, R, Python.

1. Introduction

Canadian carceral, police, and oversight data are produced by a fragmented administrative landscape: provincial corrections agencies publish placement and segregation microdata, federal Correctional Service Canada releases aggregate operational statistics under court-ordered transparency obligations, provincial oversight bodies investigate police-involved deaths and injuries, and municipal police forces release event-level open data on crime occurrences. Each

source has its own schema, unit of observation, temporal coverage, and access path. Analyses that confine themselves to a single source inherit that source’s coverage gaps; analyses that combine sources must reconcile incompatible schemas before any estimator can be applied. The result is a literature in which methodologically careful single-source work coexists with multi-source descriptive work that under-uses available estimators (Doob 2023; Zorro Medina 2023; Sprott and Doob 2023).

This paper introduces the MRM (McNamara–Ruhela–Medina) framework, a multi-source statistical foundation that brings these data streams together under a coordinated set of estimators and identification primitives. The framework’s identification strategy — staggered two-way-fixed-effects with leads-and-lags Granger diagnostics for the parallel-trends assumption, multi-source data integration over five jurisdictional records, mechanism categorisation grounded in deterrence, routine-activities, and certainty theory, and the inequality-effects-of-criminal-law framing — is established in Zorro Medina (2023). The present framework adopts that lineage and extends it to a Canadian carceral setting.

The framework is implemented as a curated set of MRM modules in MORIE (Ruhela 2026c,b), a dual-language (R and Python) toolkit released under GPL-2.0-only. The companion software papers describe the public APIs and reproducible examples; the present paper concentrates on the statistical foundations.

Three contributions are advanced. First, a unified notation for the five Canadian sources is established (Section 2) that permits a single estimator family to operate across all of them. Second, a ten-estimator per-individual ensemble is derived in its MRM instantiation (Section 3) together with a parallel aggregate χ^2 family (Section 4) and a Mandela Rules classifier (Section 5) that operates at both the federal and provincial level. The aggregate family is exercised empirically to reproduce every published χ^2 statistic from the Sprott–Doob–Iftene record to within 0.01 (Sprott and Doob 2023; Iftene and Doob 2024). Third, the framework is extended into the spatio-temporal domain through a self-exciting Hawkes process specification (Section 7) applied to Toronto Police Service open data.

A headline empirical finding is reported in Section 5: the provincial Ontario Mandela “torture”-classified proportion rises from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding by 10.7 percentage points at peak the federal SIU rate of 9.9% that Sprott and Doob (2023) report for fiscal 2019–2020. Under the broader restrictive-confinement classification advanced in Section 5, the 2025 rate reaches 40.9%.

The rest of the paper is organised as follows. Section 2 establishes the notation and the five source schemas. Section 3 derives the ten-estimator per-individual ensemble. Section 4 presents the χ^2 family for aggregate contingency tables. Section 5 introduces the Mandela Rules classifier and reports the empirical headline. Section 6 maps the deterrence / routine-activities / certainty mechanism categorisation onto the estimator family. Section 7 specifies the Hawkes self-exciting process for TPS data. Section 8 summarises the implementation in MORIE. Section 9 concludes.

2. Notation and the five Canadian sources

2.1. Notation

Let $i \in \{1, \dots, n\}$ index individuals (in OTIS) or events (in TPS), and let $t \in \{t_0, \dots, t_T\}$

index discrete time points (fiscal quarters for OTIS, calendar days for TPS, fiscal years for the aggregate sources). Each unit i at time t carries covariates $X_{it} \in \mathbb{R}^p$, a binary treatment indicator $D_{it} \in \{0, 1\}$, and an outcome $Y_{it} \in \mathbb{R}$. The treatment indicator denotes the presence of the analytical treatment under study, such as a Mandela-classified placement, a SIU Section 34 trigger, or a TPS major-crime category.

For aggregate sources, let $a \in \{1, \dots, A\}$ index categories of a contingency table and let $j \in \{1, \dots, J\}$ index a stratifying variable (e.g. jurisdiction, fiscal year, demographic group). The observed count is N_{aj} and the expected count under independence is $\hat{E}_{aj}$.

Causal estimands are defined under the standard potential-outcomes framework: $Y_{it}(d)$ denotes the outcome the unit would have realised had its treatment been set to $d \in \{0, 1\}$. The identifying assumptions of conditional exchangeability $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$, positivity $0 < \Pr(D = 1 \mid X) < 1$, and the stable-unit-treatment-value assumption are maintained throughout unless explicitly weakened.

2.2. The five sources

The five Canadian sources that the framework integrates are summarised in Table 1.

Table 1: The five Canadian carceral, police, and oversight data sources integrated by the MRM framework.

Source	Producer	Unit	Period	Access
OTIS A01-RCDD	MCSCS (Ontario)	individual–placement	2014–present	open
SIU IAP	Federal IAP	quarterly aggregate	2019–2024	open
Ontario SIU	Ontario SIU	investigation report	2009–present	open (scrape)
TPS	Toronto Police	event–day	2014–present	open
CCRSO	Public Safety CA	annual aggregate	2007–present	open

OTIS A01-RCDD is the Ontario Ministry of Community Safety and Correctional Services restrictive-confinement detailed dataset, a person-day-level record that includes age band, gender, alert flags, segregation type, and exit code. It is the only source in the table that supports per-individual estimands directly. The schema is published with each annual release.

SIU IAP comprises the four quarterly reports of the federal Structured Intervention Unit Implementation Advisory Panel (2019–2022) and the parallel Sprott–Doob–Iftene academic replication corpus (Sprott and Doob 2023; Iftene and Doob 2024), which includes contingency tables that report the federal Mandela rate of 9.9% used as the reference point in Section 5.

Ontario SIU comprises the case-disposition reports published by the Ontario Special Investigations Unit at siu.on.ca. The framework’s data pipeline ingests these reports through a maintained scraper released under the GPL-2 licence of MORIE (Ruhela 2026c). The corpus covers the SIU’s operational history.

TPS open data is the Toronto Police Service’s daily event-level release of crime occurrences with spatial coordinates aggregated to neighbourhood, supporting the spatio-temporal Hawkes specification of Section 7.

CCRSO, the Corrections and Conditional Release Statistical Overview, is the principal federal aggregate release. It enters the record through its tabling under the *Federal Court of Canada*

docket T-539-20 (the affidavit-based admissibility process referred to in [Doob 2023](#)) and provides the federal denominator for several of the framework’s aggregate χ^2 statistics.

2.3. Time-alignment and the panel structure

The five sources do not share a fiscal calendar. OTIS uses the Ontario fiscal year (April–March); SIU IAP the federal fiscal year; TPS the calendar year; and CCRSO the federal fiscal year. The framework reconciles these through a canonical fiscal quarter index implemented as an interval map in MORIE’s `align_fiscal()` routine ([Ruhela 2026c](#)).

3. Per-individual estimators: the MRM ensemble

The per-individual ensemble specifies ten estimators of the average treatment effect (ATE) and the average treatment effect on the treated (ATT) under the identifying assumptions stated in Section 2. The estimators are grouped into five families.

3.1. Inverse-probability-weighting estimators

The first family is built around the propensity score $e(X) = \Pr(D = 1 \mid X)$. The Horvitz–Thompson IPW estimator of the ATE is

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i Y_i}{\hat{e}(X_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{e}(X_i)} \right], \quad (1)$$

and the stabilised IPW (sIPW) variant is

$$\hat{\tau}_{\text{sIPW}} = \frac{\sum_i D_i Y_i / \hat{e}(X_i)}{\sum_i D_i / \hat{e}(X_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - D_i) / (1 - \hat{e}(X_i))}. \quad (2)$$

The framework follows [Cole and Hernán \(2008\)](#) and trims extreme weights at the $[0.01, 0.99]$ quantiles of $\hat{e}$ to reduce variance inflation at the cost of a small finite-sample bias.

The ATT analogue,

$$\hat{\tau}_{\text{IPW}}^{\text{ATT}} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_1} \sum_{i:D_i=0} \frac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} Y_i, \quad (3)$$

follows [Imbens and Rubin \(2015\)](#).

3.2. Augmented inverse-probability weighting

The doubly-robust AIPW estimator combines the propensity score with a working outcome regression model $\mu_d(X) = \mathbb{E}[Y \mid X, D = d]$. The estimator is

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]. \quad (4)$$

AIPW is consistent when either e or μ_d (but not necessarily both) is consistently estimated; this is the double-robustness property emphasised in [Robins *et al.* \(1994\)](#) and [Bang and Robins \(2005\)](#). The framework follows the implementation conventions of [Funk *et al.* \(2011\)](#).

3.3. Double-machine-learning interactive regression

The interactive regression model (IRM) of Chernozhukov *et al.* (2018) relaxes the parametric assumptions on μ_d by allowing it to be estimated with arbitrary machine-learning methods, provided that the resulting nuisance estimators converge at sufficient rates for the Neyman-orthogonal score function to remain locally insensitive to nuisance error. The MRM instantiation wraps the R package **DoubleML** (Bach *et al.* 2024) via its **DoubleMLIRM** class with the **mlr3** learner ecosystem (Lang *et al.* 2019), and the Python package **DoubleML** via its **DoubleMLIRM** class with the **scikit-learn** learner interface (Bach *et al.* 2022; Pedregosa *et al.* 2011). The MRM contributions are (i) the choice of sample-splitting fold count for OTIS (five folds, two repetitions) calibrated to the panel dimensions of A01-RCDD, and (ii) a wrapper convention that exposes both ATE and ATT in a single call.

3.4. Propensity-score matching

Matching estimators construct a counter-factual sample by pairing each treated unit with one or more untreated units whose propensity score is within a caliper of the treated unit's. The framework implements one-to- k caliper matching with replacement following Stuart (2010), with default $k = 4$ and caliper width $0.2 \times \text{SD}(\text{logit } e)$ per Austin (2011). The matched-sample ATT is

$$\hat{\tau}_{\text{match}}^{\text{ATT}} = \frac{1}{n_1} \sum_{i:D_i=1} \left[Y_i - \frac{1}{k} \sum_{j \in \mathcal{M}(i)} Y_j \right], \quad (5)$$

where $\mathcal{M}(i)$ denotes the matched-control set for treated unit i .

3.5. Partial-linear and super-learner variants

The partial-linear regression model (PLR) of Chernozhukov *et al.* (2018) specifies $Y_i = D_i\theta_0 + g_0(X_i) + U_i$, with $\mathbb{E}[U_i | X_i, D_i] = 0$, and recovers θ_0 through orthogonalisation against $\hat{g}_0$. The super-learner variant of van der Laan *et al.* (2007) replaces the single nuisance estimator with a cross-validated convex combination of candidate learners. The MRM instantiation uses **glmnet** (Friedman *et al.* 2010) and **ranger** (Wright and Ziegler 2017) as the candidate library for both $\hat{g}_0$ and $\hat{m}_0$, with weights chosen by ten-fold cross-validation on the in-sample squared-error loss.

3.6. Sensitivity analysis

Each estimand is reported with an E-value (VanderWeele and Ding 2017), the minimum unmeasured-confounder-outcome and unmeasured-confounder-treatment association required to explain away the observed effect under the worst-case unmeasured-confounder distribution. For OTIS estimands with $\hat{\tau}$ corresponding to an odds ratio of $\widehat{\text{OR}}$, the E-value is

$$\text{E-value} = \widehat{\text{OR}} + \sqrt{\widehat{\text{OR}}(\widehat{\text{OR}} - 1)}. \quad (6)$$

Values exceeding 2.0 are conventionally interpreted as robust to plausible unmeasured confounding by clinical-severity covariates that are not available in the open A01-RCDD record.

3.7. Generalised-estimating-equation clustering

When the unit of analysis is the placement-record (a single individual may contribute many

records over a fiscal year) the within-person correlation between observations must be absorbed. The MRM clustering grid extends the per-individual estimators with a generalised-estimating-equation (GEE) clustering step over four candidate cluster keys: institution, regional cluster, individual, and a combined institution–quarter key. Each candidate yields a sandwich-variance estimate; the framework reports all four to allow sensitivity-to-clustering inspection ([Hubbard et al. 2010](#); [Cameron and Miller 2015](#)).

4. Aggregate estimators: the χ^2 family

For aggregate sources (SIU IAP, CCRSO) the unit of observation is a count, not an individual. The MRM aggregate family specifies χ^2 statistics under various marginal-fixing conventions to recover the contingency-table comparisons reported in the published Sprott–Doob–Iftene record.

For an $A \times J$ table with observed cell count N_{aj} , total $N_{..}$, row total $N_{a.}$, and column total $N_{.j}$, the standard χ^2 statistic under the independence null is

$$\chi^2 = \sum_{a=1}^A \sum_{j=1}^J \frac{(N_{aj} - \hat{E}_{aj})^2}{\hat{E}_{aj}}, \quad \hat{E}_{aj} = \frac{N_{a.}N_{.j}}{N_{..}}. \quad (7)$$

The framework’s χ^2 family includes (i) the standard statistic, (ii) the Yates-continuity-corrected statistic for 2×2 subtables, (iii) the Cochran–Armitage trend test for ordered A , and (iv) the Mantel–Haenszel statistic for stratified tables. Empirical replication of every published Sprott–Doob–Iftene χ^2 to within 0.01 is demonstrated in the MORIE test suite ([Ruhela 2026c](#)).

5. Mandela Rules classifier and the headline finding

The United Nations Standard Minimum Rules for the Treatment of Prisoners (the Mandela Rules) define a prisoner as held in prolonged solitary confinement if a single continuous placement exceeds fifteen days ([United Nations General Assembly 2015](#)). The MRM Mandela Rules classifier operationalises this definition at both the federal and provincial level.

5.1. Federal operationalisation

At the federal level, the Sprott–Doob–Iftene record ([Sprott and Doob 2023](#)) classifies a Section 34 Structured Intervention Unit placement as Mandela-torture when the recorded placement duration exceeds fifteen days and one or more of the “meaningful contact” criteria is unmet. Their published rate for fiscal 2019–2020 is 9.9% of all Section 34 placements. The MRM implementation replicates this classification logic on the same data and recovers 9.9% to within rounding.

5.2. Provincial operationalisation

At the provincial level, OTIS A01-RCDD does not include the direct “meaningful contact” fields available in the federal data. The framework therefore operationalises the Mandela classification on duration alone, applied to the segregation-type placements recorded in A01-RCDD. The conservative rate uses the strict >15-day duration threshold; the broader

restrictive-confinement rate uses the union of segregation-type placements and high-alert placements satisfying the same duration threshold.

5.3. Cross-jurisdiction comparison

Table 2 reports the federal benchmark and the provincial Mandela rate over fiscal 2023–2025.

Table 2: Mandela Rules classification rates, federal benchmark and provincial Ontario, fiscal 2019–2025. Federal rate from [Sprott and Doob \(2023\)](#); provincial rates computed from OTIS A01-RCDD using the MRM classifier ([Ruhela 2026c,b](#)).

Jurisdiction (period)	Mandela rate	Broader RC rate	Source
Federal SIU 2019–2020	9.9%	—	Sprott and Doob 2023
Provincial Ontario 2023	12.5%	—	MRM
Provincial Ontario 2024	16.5%	—	MRM
Provincial Ontario 2025	20.6%	40.9%	MRM

The provincial Mandela rate exceeds the federal benchmark in every fiscal year for which both are available, and by 2025 the gap reaches 10.7 percentage points. Under the broader restrictive-confinement classification — which counts high-alert placements alongside segregation-type placements — the 2025 provincial rate is 40.9%. The implication for the ongoing *Jahn*-settlement oversight framework is that the duration-and-condition-based monitoring metrics currently reported do not fully capture the prevalence of placements that would satisfy the Mandela definition of prolonged solitary confinement.

5.4. Caveats

Three caveats apply. First, the provincial classifier operates on duration alone; absent “meaningful contact” criteria, the classification is necessarily duration-conservative and may under-count placements that would qualify on additional criteria. Second, OTIS A01-RCDD identifiers are reassigned at fiscal boundaries; within-person placement chaining is therefore restricted to single-fiscal-year windows. Third, the broader-restrictive-confinement rate uses a union rule (segregation-type OR high-alert) that the Sprott–Doob–Iftene record does not separately validate; readers who require strict replication should restrict attention to the Mandela-classified row.

6. Mechanism categorisation

Each estimator in the per-individual ensemble of Section 3 can be tied to one or more substantive sociolegal mechanisms following the categorisation of [Zorro Medina \(2023\)](#) and its precursors in [Becker \(1968\)](#), [Cohen and Felson \(1979\)](#), and [Apel \(2013\)](#). The framework distinguishes three mechanism classes.

Deterrence. The deterrence mechanism applies when the treatment changes the expected utility of an action by altering its perceived cost. Estimands under deterrence are typically interpreted as the effect of a sentencing or oversight regime on the rate of subsequent be-

haviour. The IPW and AIPW estimators of Section 3.1 and Section 3.2 provide the natural decomposition: $\hat{\tau}_{\text{AIPW}}$ separates the regression component (the deterrent effect transmitted through expected outcomes) from the weighting component (the deterrent effect transmitted through selection-into-treatment).

Routine activities. The routine-activities mechanism applies when the treatment changes the convergence in space and time of motivated offenders, suitable targets, and capable guardians (Cohen and Felson 1979). Estimands under this mechanism are appropriately specified at the neighbourhood-day level (the Hawkes specification of Section 7) rather than the individual-placement level.

Certainty. The certainty mechanism applies when the treatment changes the perceived probability of detection or sanction rather than its magnitude. The propensity-score matching and partial-linear estimators of Section 3 provide the estimand: by matching on the covariates that drive selection into treatment, the matched comparison estimates the effect under a common conditional probability of treatment given covariates.

7. Hawkes self-exciting process for TPS events

The Hawkes process (Hawkes 1971; Ogata 1981) provides a parsimonious generative model for event sequences with temporal clustering. The conditional intensity of a one-dimensional Hawkes process is

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} g(t - t_i), \quad (8)$$

where $\mu > 0$ is the baseline intensity, $\mathcal{H}_t$ is the history of events up to time t , and $g(\cdot) \geq 0$ is the triggering kernel. The branching ratio $n = \int_0^\infty g(s) ds$ satisfies $n < 1$ for a stationary process.

7.1. Markovian (exponential) kernel

The classical Markovian Hawkes specification fixes $g(s) = \alpha\beta e^{-\beta s}$, yielding the conditional intensity

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} \beta e^{-\beta(t-t_i)}, \quad n = \alpha, \quad (9)$$

which permits a recursive computation of the intensity through the augmented state $S(t) = \sum_{t_i < t} \beta e^{-\beta(t-t_i)}$ satisfying $dS/dt = -\beta S$ between events with a unit jump at each event time (Ogata 1981; Daley and Vere-Jones 2003).

7.2. Non-Markovian (power-law) kernel

For crime-event data the empirical autocorrelation function decays more slowly than exponentially and a power-law triggering kernel is appropriate (Mohler *et al.* 2011):

$$g(s) = \alpha(s + c)^{-(1+\omega)}, \quad \omega > 0. \quad (10)$$

The MRM-Hawkes specification fits both kernels by maximum likelihood and selects between them using the Bayes information criterion.

7.3. Spatio-temporal extension

For TPS events with spatial coordinates $\mathbf{x}_i \in \mathbb{R}^2$, the spatio-temporal Hawkes intensity is

$$\lambda(t, \mathbf{x} \mid \mathcal{H}_t) = \mu(\mathbf{x}) + \sum_{t_i < t} g(t - t_i) f(\mathbf{x} - \mathbf{x}_i), \quad (11)$$

where f is a spatial-displacement density (typically a Gaussian or a Student- t). Applied to TPS open data, the spatio-temporal specification recovers daily-neighbourhood clustering structure that descriptive analyses do not.

7.4. Implementation

The MRM-Hawkes implementation in MORIE provides both Markovian and power-law fits via maximum likelihood, the Ogata thinning algorithm for simulation, and a branching-ratio diagnostic for stationarity (Ruhela 2026a,c,b). Goodness-of-fit is assessed through the time-rescaling theorem of Brown *et al.* (2002): under the fitted model the rescaled inter-event times should be exponentially distributed; a Kolmogorov–Smirnov test against the unit-rate exponential provides the goodness-of-fit p -value.

8. Implementation in MORIE

The framework is implemented as a curated set of MRM modules in MORIE, a dual-language (R and Python) toolkit released under GPL-2.0-only. The companion software papers (Ruhela 2026c,b) describe the public APIs and reproducible examples in detail; the present section documents only the interface between the framework’s mathematical specification and its software realisation.

The ten per-individual estimators of Section 3 are exposed as `morie::estimate_att()`, `morie::estimate_atc()`, `morie::estimate_aipw()`, and `morie::estimate_irm()` (and the corresponding Python `morie.causal.*` functions). The aggregate χ^2 family of Section 4 is exposed as `morie::chi2_doob()`. The Mandela Rules classifier of Section 5 is exposed as `morie::classify_mandela()`. The Hawkes specification of Section 7 is exposed as `morie::fit_hawkes()` with explicit `kernel = "exponential"` and `kernel = "power_law"` options. The framework’s software DOI is Ruhela (2026c).

9. Conclusion

The MRM framework brings five Canadian carceral, police, and oversight data streams together under a coordinated set of estimators and identification primitives. The ten-estimator per-individual ensemble, the aggregate χ^2 family, the Mandela Rules classifier, the mechanism categorisation, and the Hawkes self-exciting specification collectively provide the statistical foundation under which empirical analyses of these sources may proceed with a single notation and a single estimator family. The framework’s empirical reach is demonstrated by the replication of every published Sprott–Doob–Iftene χ^2 to within 0.01 and by the headline finding that the provincial Ontario Mandela rate now exceeds the federal benchmark by 10.7 percentage points at peak.

The framework’s open-source realisation in the MORIE toolkit ([Ruhela 2026c,b](#)) makes every estimator and every Mandela classification reproducible on the open Ontario and federal data records. Extensions of the framework to additional administrative sources (e.g. the Royal Canadian Mounted Police annual statistics, the Public Health Agency of Canada substance-use surveys) follow the same notational and mechanism-categorisation discipline.

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